

# Week 3 - Practice Questions (Individual Sets)

Hi everyone,

Try to do these questions on your own first. If it feels difficult, don't worry - we will solve them together in class.

**How to use this sheet:** Do the questions in your assigned set. Write the steps in your notebook first, then code in Java. Use the test cases to verify your output.

## Vishnu

You have 5 questions. For each one: read the task, follow the steps, then test using the inputs below.

### 1. Even or Odd

**Task:** Given a number, print Even or Odd.

**Steps to solve (algorithm):**

- 1) Take the number from the user.
- 2) Check if it divides by 2 with remainder 0.
- 3) If remainder is 0 print Even, otherwise print Odd.

**Sample**

Input:

7

Output:

Odd

| Test case input | Expected output |
|-----------------|-----------------|
| 10              | Even            |
| 7               | Odd             |
| 0               | Even            |

### 2. Sum of digits

**Task:** Print the sum of all digits in the number.

**Steps to solve (algorithm):**

- 1) Take the number from the user.
- 2) Take the last digit and add it to the sum.
- 3) Remove the last digit from the number.
- 4) Repeat until no digits remain.
- 5) Print the final sum.

**Sample**

Input:

123

Output:

6

| Test case input | Expected output |
|-----------------|-----------------|
| 123             | 6               |
| 908             | 17              |
| 0               | 0               |

### 3. Largest of 3 numbers

**Task:** Read 3 numbers and print the largest.

**Steps to solve (algorithm):**

- 1) Take three numbers from the user.
- 2) Compare them to find which one is the greatest.
- 3) Print the greatest number.

**Sample**

Input:

3 9 2

Output:

9

| Test case input | Expected output |
|-----------------|-----------------|
| 3 9 2           | 9               |
| 5 5 1           | 5               |
| -1 -6 -3        | -1              |

### 4. Multiplication Table

**Task:** Print the multiplication table of a number from 1 to 10.

**Steps to solve (algorithm):**

- 1) Take a number n from the user.
- 2) Multiply n by 1, 2, 3 ... up to 10.
- 3) Print each line in the format: n x i = result.

**Sample**

Input:

2

Output:

2 x 1 = 2 ... 2 x 10 = 20

| Test case input | Expected output   |
|-----------------|-------------------|
| 2               | ... 2 x 10 = 20   |
| 5               | ... 5 x 10 = 50   |
| -3              | ... -3 x 10 = -30 |

## 5. Palindrome number

**Task:** Print Palindrome if the number reads the same forward and backward.

### Steps to solve (algorithm):

- 1) Take the number and keep it as the original.
- 2) Reverse the number.
- 3) Compare the original and the reversed number.
- 4) If they are the same, print Palindrome, else print Not Palindrome.

### Sample

Input:

121

Output:

Palindrome

| Test case input | Expected output |
|-----------------|-----------------|
| 121             | Palindrome      |
| 123             | Not Palindrome  |
| 0               | Palindrome      |

# Naveen

You have 5 questions. For each one: read the task, follow the steps, then test using the inputs below.

## 1. Prime number check

**Task:** Print Prime or Not Prime.

**Steps to solve (algorithm):**

- 1) Take the number n.
- 2) If n is less than or equal to 1, print Not Prime.
- 3) Try dividing n by smaller numbers starting from 2.
- 4) If any division gives remainder 0, print Not Prime.
- 5) If none divide completely, print Prime.

**Sample**

Input:

7

Output:

Prime

| Test case input | Expected output |
|-----------------|-----------------|
| 7               | Prime           |
| 12              | Not Prime       |
| 1               | Not Prime       |

## 2. Factorial

**Task:** Print factorial of n ( $1 \times 2 \times \dots \times n$ ).

**Steps to solve (algorithm):**

- 1) Take the number n.
- 2) Multiply all numbers from 1 up to n.
- 3) Print the final multiplied value.

**Sample**

Input:

5

Output:

120

| Test case input | Expected output |
|-----------------|-----------------|
| 5               | 120             |
| 0               | 1               |
| 7               | 5040            |

## 3. Count digits

**Task:** Print how many digits are in the number.

**Steps to solve (algorithm):**

- 1) Take the number.
- 2) If the number is 0, answer is 1.
- 3) Remove one digit at a time from the end and count how many times you remove.
- 4) When all digits are removed, print the count.

**Sample**

Input:

999

Output:

3

| Test case input | Expected output |
|-----------------|-----------------|
| 12345           | 5               |
| 0               | 1               |
| -98             | 2               |

#### 4. Fibonacci (first N terms)

**Task:** Print first n Fibonacci numbers.

**Steps to solve (algorithm):**

- 1) Take the number n (how many terms).
- 2) Start the sequence with 0 and 1.
- 3) Print terms one by one.
- 4) Each next term is the sum of the previous two.
- 5) Stop after printing n terms.

**Sample**

Input:

5

Output:

0 1 1 2 3

| Test case input | Expected output |
|-----------------|-----------------|
| 1               | 0               |
| 5               | 0 1 1 2 3       |
| 7               | 0 1 1 2 3 5 8   |

#### 5. Leap Year

**Task:** Print Leap Year or Not Leap Year for a given year.

**Steps to solve (algorithm):**

- 1) Take the year.
- 2) If divisible by 400, it is a Leap Year.
- 3) Else if divisible by 100, it is Not a Leap Year.

- 4) Else if divisible by 4, it is a Leap Year.
- 5) Else it is Not a Leap Year.

**Sample**

Input:

2024

Output:

Leap Year

| Test case input | Expected output |
|-----------------|-----------------|
| 2000            | Leap Year       |
| 1900            | Not Leap Year   |
| 2024            | Leap Year       |

## Charan

You have 5 questions. For each one: read the task, follow the steps, then test using the inputs below.

### 1. Reverse a number

**Task:** Reverse the digits of the number.

**Steps to solve (algorithm):**

- 1) Take the number from the user.
- 2) Take the last digit and build a new number with it.
- 3) Remove the last digit from the original number.
- 4) Repeat until no digits remain.
- 5) Print the reversed number.

**Sample**

Input:

123

Output:

321

| Test case input | Expected output |
|-----------------|-----------------|
| 123             | 321             |
| 1000            | 1               |
| 908             | 809             |

### 2. Print even numbers (1 to N)

**Task:** Print all even numbers from 1 to n.

**Steps to solve (algorithm):**

- 1) Take the number n.
- 2) Check each number from 1 to n.
- 3) Print only the numbers divisible by 2.

**Sample**

Input:

10

Output:

2 4 6 8 10

| Test case input | Expected output |
|-----------------|-----------------|
| 10              | 2 4 6 8 10      |
| 1               | (no output)     |
| 7               | 2 4 6           |

### 3. Electricity usage slab

**Task:** Given units, print Low Usage (<100), Medium Usage (<300), else High Usage.

**Steps to solve (algorithm):**

- 1) Take the units.
- 2) If units are less than 100, print Low Usage.
- 3) Else if units are less than 300, print Medium Usage.
- 4) Else print High Usage.

**Sample**

Input:

250

Output:

Medium Usage

| Test case input | Expected output |
|-----------------|-----------------|
| 80              | Low Usage       |
| 250             | Medium Usage    |
| 400             | High Usage      |

#### 4. Vowel count

**Task:** Count vowels (a, e, i, o, u) in the given text.

**Steps to solve (algorithm):**

- 1) Take the word or line of text.
- 2) Check each character one by one.
- 3) If the character is a vowel, increase the count.
- 4) Print the final count.

**Sample**

Input:

Reshma

Output:

2

| Test case input | Expected output |
|-----------------|-----------------|
| HELLO           | 2               |
| sky             | 0               |
| education       | 5               |

#### 5. Simple menu

**Task:** Input choice: 1 Tea, 2 Coffee, 3 Juice; otherwise Invalid choice.

**Steps to solve (algorithm):**

- 1) Take the choice number.
- 2) If choice is 1 print Tea.
- 3) If choice is 2 print Coffee.
- 4) If choice is 3 print Juice.



5) Otherwise print Invalid choice.

**Sample**

Input:

2

Output:

Coffee

| Test case input | Expected output |
|-----------------|-----------------|
| 1               | Tea             |
| 3               | Juice           |
| 5               | Invalid choice  |

## Megha

You have 5 questions. For each one: read the task, follow the steps, then test using the inputs below.

### 1. Armstrong number (3-digit)

**Task:** Print Armstrong or Not Armstrong for a 3-digit number.

**Steps to solve (algorithm):**

- 1) Take a 3-digit number.
- 2) Separate the digits.
- 3) Cube each digit (digit x digit x digit) and add them.
- 4) If the sum equals the original number, print Armstrong.
- 5) Otherwise print Not Armstrong.

**Sample**

Input:

153

Output:

Armstrong

| Test case input | Expected output |
|-----------------|-----------------|
| 153             | Armstrong       |
| 370             | Armstrong       |
| 123             | Not Armstrong   |

### 2. Largest digit in a number

**Task:** Print the largest digit present in the number.

**Steps to solve (algorithm):**

- 1) Take the number.
- 2) Check digits one by one.
- 3) Keep track of the biggest digit you see.
- 4) After checking all digits, print the biggest digit.

**Sample**

Input:

3942

Output:

9

| Test case input | Expected output |
|-----------------|-----------------|
| 3942            | 9               |
| 0               | 0               |
| 777             | 7               |

### 3. Discount applied (if-only)

**Task:** If bill amount is 1000 or more, print Discount Applied. Otherwise print nothing.

**Steps to solve (algorithm):**

- 1) Take the bill amount.
- 2) If amount is 1000 or more, print Discount Applied.
- 3) Otherwise do not print anything.

**Sample**

Input:

1200

Output:

Discount Applied

| Test case input | Expected output  |
|-----------------|------------------|
| 1200            | Discount Applied |
| 500             | (no output)      |
| 1000            | Discount Applied |

#### 4. Login + OTP (nested if)

**Task:** If username is user, then OTP must be 1111 to login.

**Steps to solve (algorithm):**

- 1) Take username.
- 2) If username matches required name, then take OTP.
- 3) If OTP matches, print Login Success; otherwise print Wrong OTP.
- 4) If username does not match, print Invalid Username.

**Sample**

Input:

user then 1111

Output:

Login Success

| Test case input | Expected output  |
|-----------------|------------------|
| user + 1111     | Login Success    |
| user + 9999     | Wrong OTP        |
| abc + 1111      | Invalid Username |

#### 5. Vote eligibility

**Task:** If age is 18 or more, print Eligible. Otherwise print Not Eligible.

**Steps to solve (algorithm):**

- 1) Take age.
- 2) If age is 18 or more, print Eligible.
- 3) Otherwise print Not Eligible.

**Sample**

Input:

16

Output:

Not Eligible

| Test case input | Expected output |
|-----------------|-----------------|
| 21              | Eligible        |
| 18              | Eligible        |
| 16              | Not Eligible    |